GENERATOR [i-ELOOP]

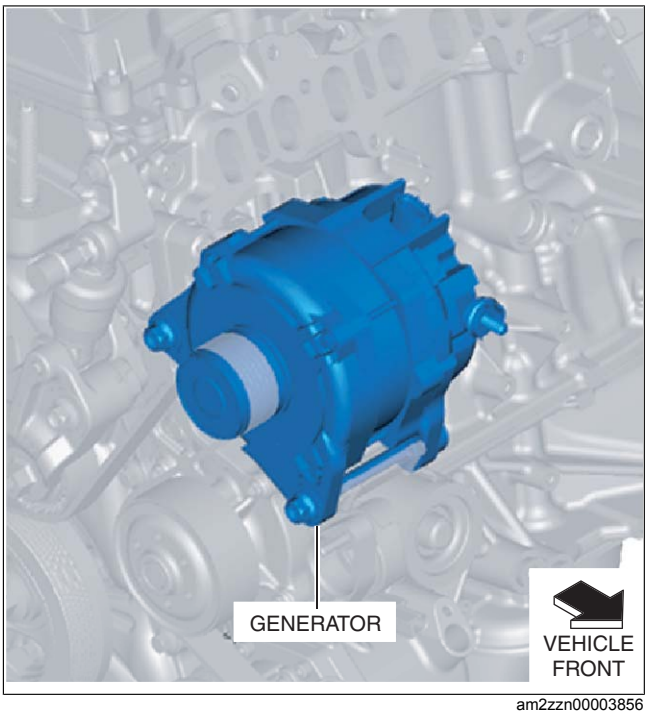
id131704501000

Purpose, Function

- A variable voltage type generator has been adopted for the generator which can generate 12—24.3 V according to the capacitor (i-ELOOP) storage conditions.

Construction

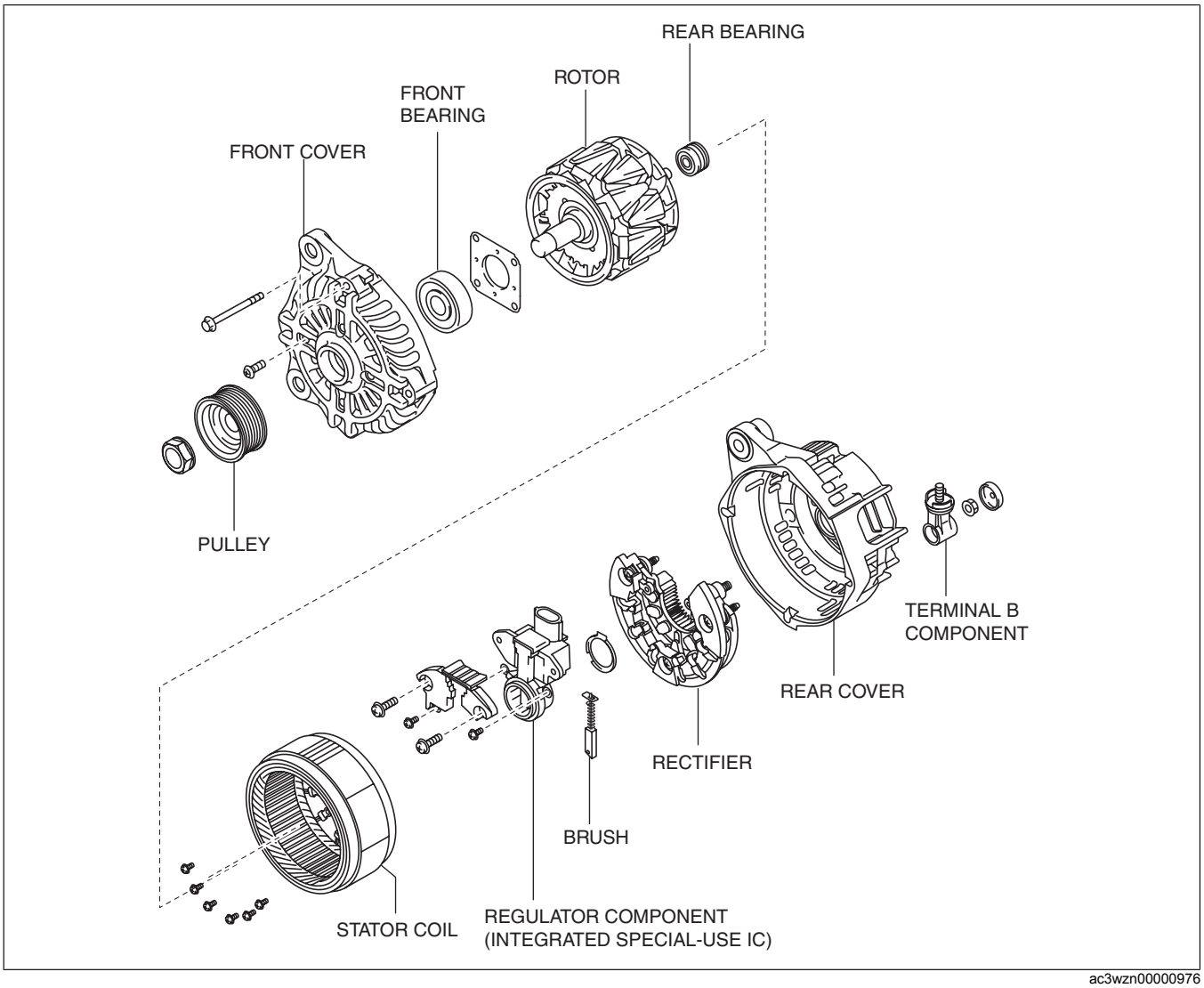
- The generator is located on the left front side of the engine.



am2zzn00003856

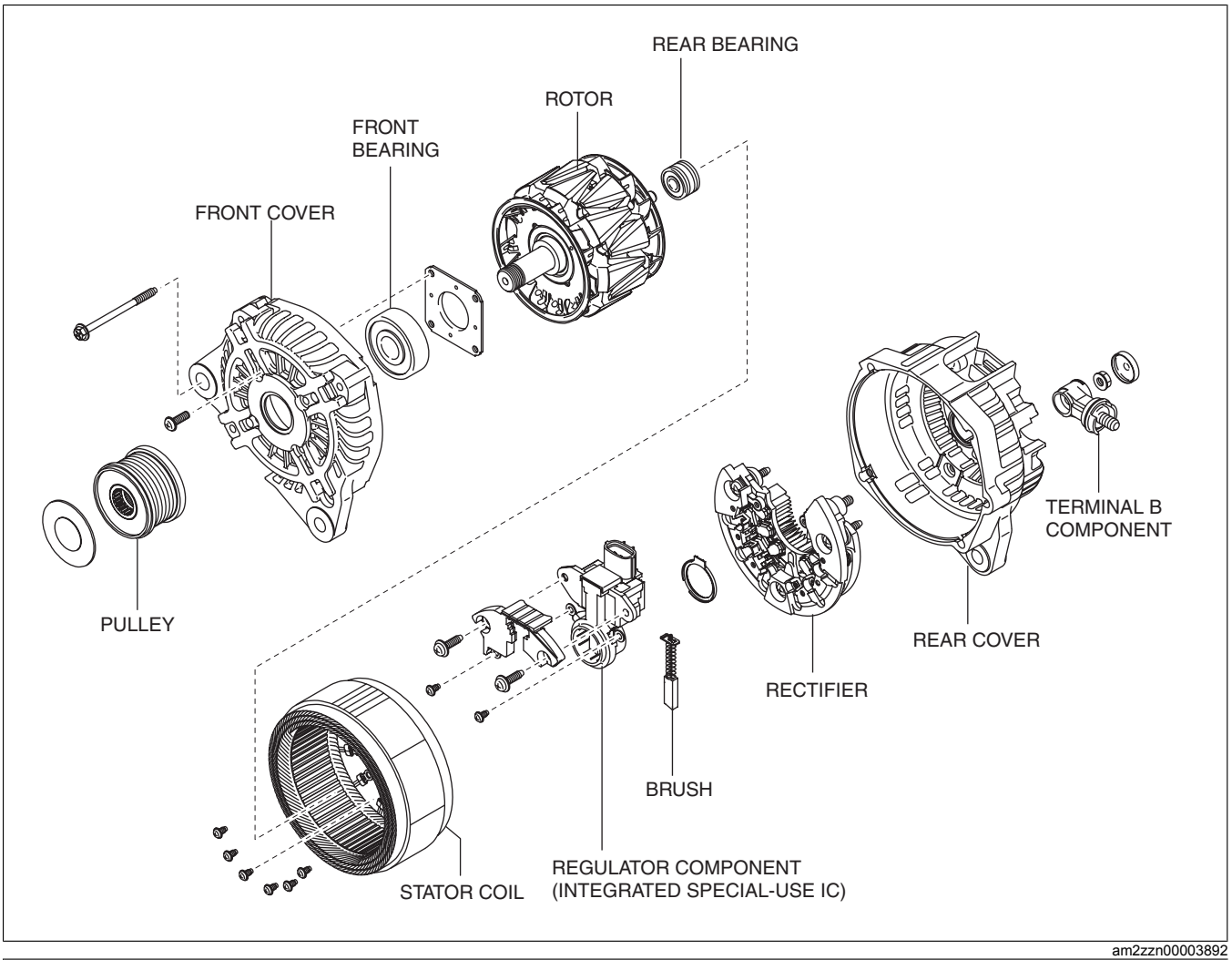
- The generator consists of the following parts:

SKYACTIV-G 1.5

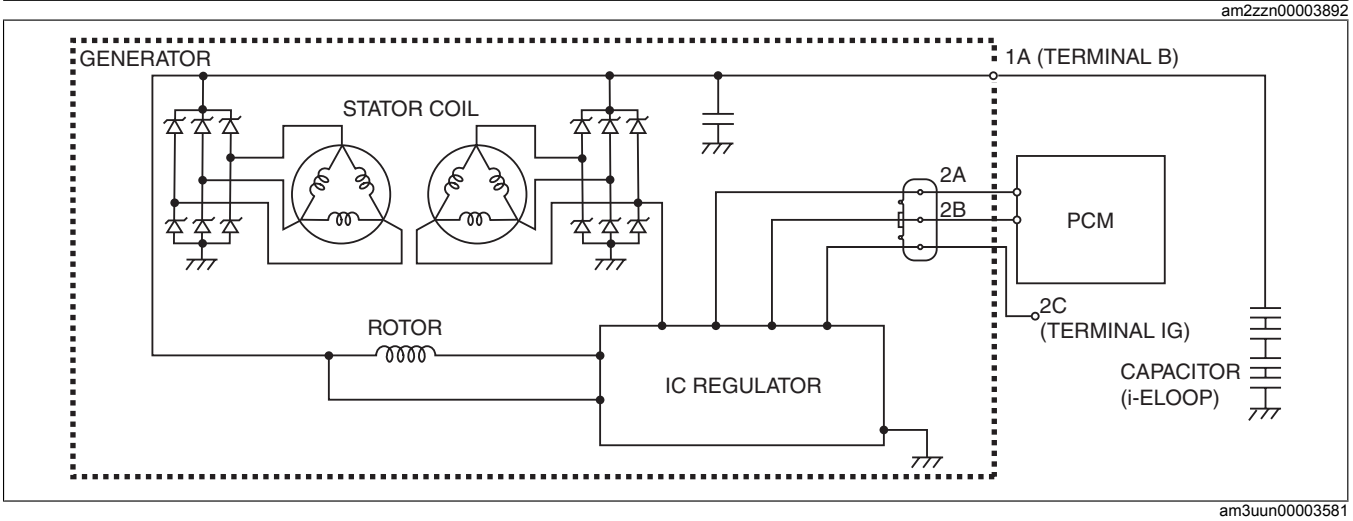


ac3wzn00000976

SKYACTIV-D 1.5



am2zzn00003892



am3uun00003581

- To handle a maximum of 24.3 V, the voltage resistance performance of the diode has been improved.
- The IC regulator built into the regulator component has the following protection functions which function during overheating/over-voltage/over-current.

  Excess temperature protection function

  — If the IC regulator reaches the specified temperature, the generator output current is restricted to suppress a rise in the temperature of the generator.

  Excess voltage protection function

  — If an excess excitation current control signal from the PCM is input, the generator output voltage is restricted to suppress application of excess voltage to the capacitor (i-ELOOP).

  Excess-current protection function

— If the excitation current reaches the specified value, the excitation current is interrupted to restrict power generation by the generator.

- The initial excitation current to the rotors is supplied from the capacitor (i-ELOOP). When excitation current is supplied from the capacitor (i-ELOOP) and the generator begins generating electricity, the excitation current from the capacitor (i-ELOOP) is interrupted. After the generator begins generating electricity, the power generated by the generator is used for excitation current.

Fail-safe

DTC	Fail-safe
P0A12:00	• Inhibits engine-stop by operating the i-stop function.
P2502:00	
P2503:00	
P2504:00	
P1794:00	• Inhibits engine-stop by operating the i-stop function. (With i-stop) • Inhibits the i-ELOOP control. • Switches to bypass mode which shuts the capacitor (i-ELOOP) off.